

Akshat Srivastava

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[LinkedIn](#) | [Github](#) | [Leetcode](#)

EDUCATION

Vellore Institute of Technology

Computer Science and Engineering; GPA: 8.98

Bhopal, India

St. Joseph's College

Class XII Percentage (81.7%)

Prayagraj, India

St. Joseph's College

Class X Percentage (93.8%)

Prayagraj, India

SKILLS SUMMARY

- **Languages:** Python, C++, JAVA, HTML, CSS, JavaScript
- **Framework & AI/ML:** RAG Pipeline, LLM Orchestration, React, Spring, SpringBoot
- **Cloud & MLOps:** AWS (SageMaker, IoT Core, Kinesis, S3), FinOps, MQTT, mTLS
- **Databases:** MongoDB, MySQL
- **Concepts and Architecture:** RestfulAPIs
- **Platforms and Tools:** Visual Studio Code, BlueJ, Vercel, Netlify, Git, GitHub

RELEVANT EXPERIENCE

Software Engineering Intern – Aritrak Technologies

June 2026

Tech Stack: Python, AWS (Lambda, SES, EventBridge), SaaS Architecture, Offline RAG Pipelines, LLM Orchestration, LoRA/QLoRA, Ollama (Qwen, Llama, Mistral), Git

- Engineered and deployed an automated backend pipeline utilizing AWS Lambda, Amazon SES, and Amazon EventBridge to manage programmatically driven communications and event-driven scheduling.
- Architected and developed a scalable, AI-driven SaaS platform that integrates a custom offline RAG model to facilitate secure, localized, and private data processing.
- Fine-Tuned open-source LLMs (Qwen, Llama, and Mistral) using LoRA/QLoRA parameter-efficient techniques, deployed for high-performance execution on local devices via Ollama.

PUBLICATION

Network Coding for Efficient Data Transmission in Covid-19 Genomic Research

March 2025

- Published at the 2025 International Conference on Intelligent Control, Computing and Communications (IC3), IEEE.
- Implemented network coding techniques to enhance data transmission efficiency in protein folding simulations, achieving a consistent 50% throughput improvement under high packet loss (10%–30%).

PROJECTS

Real-Time Predictive Maintenance Data Pipeline | [LINK](#)

February 2026

Tech Stack: Python, AWS IoT Core, AWS S3, Amazon SageMaker, FinOps

- Engineered AWS IIoT pipeline (IoT Core, S3) handling 100+ telemetry events/sec.
- Trained a SageMaker model achieving high-accuracy detection for machine failure prediction.
- Achieved <1s latency for real-time anomaly detection using a custom Python subscriber; optimized cloud spend by 90% via automated FinOps resource lifecycle management.

Sahayak-AI – AI-Powered IT Helpdesk Platform | [LINK](#)

August 2026

Tech Stack: Java, Spring Boot, Spring Security, React, MySQL/H2, AWS S3, Groq (LLM Inference), JWT

- Built a full-stack, role-based (User/Agent/Admin) ticketing system with JWT authentication and server-side, department-scoped authorization.
- Integrated async Groq LLM inference to auto-classify ticket category and priority in ~150ms, with defensive fallback handling for zero downtime on AI provider failures.
- Enabled direct browser-to-S3 file uploads via pre-signed URLs, keeping the storage bucket fully private.

JeevanChakra – Personalized Healthcare Web App (Health Hackathon, Johns Hopkins University) | [LINK](#)

April 2025

Tech Stack: React, Python (ResttAPI), MongoDB, Gemini API, Speech-to-Text

- Developed a full-stack healthcare solution integrating AI-driven medical analysis, increasing self-diagnosis accuracy by 45%.
- Implemented a multi-channel reminder system for medication and appointment adherence.
- Added a voice-enabled AI chatbot, improving diagnostic speed by 30%.

CERTIFICATES

AWS Certified AI Practitioner – Foundational

May 2026

AWS Certified Machine Learning Engineer – Associate

March 2026

Applied Machine Learning in Python

December 2025